(c) Conversion of weeks to days.

We multiply weeks by 7 to convert into days.

Example 1: **Convert 5 weeks into days.**

Solution: 5 weeks = 5 x 7 days = 35 days

Example 2: **Convert 2 weeks 4 days into days.**

Solution: 2 weeks 4 days = 2 x 7 days + 4 days = 18 days

EXERCISE 5.8

(1) Convert the following into months.

(i) **5 years** (ii) **8 years 6 months**

(iii) **4 years 9 months** (iv) **10 years 2 months**

(v) **15 years 8 months** (vi) **20 years 10 months**

(2) Convert the following into days.

(i) **3 months** (ii) **3 months 12 days**

(iii) **8 months 20 days** (iv) **4 months 25 days**

(v) **10 months 28 days** (vi) **2 months 15 days**

(3) Convert the following into days.

(i) **8 weeks** (ii) **25 weeks 3 days**

(iii) **17 weeks 6 days** (iv) **30 weeks**

(v) **35 weeks 5days** (vi) **41 weeks 2 days**

Addition and subtraction of units of time

Example 1: Add 15 minutes 35 seconds and 30 minutes 20 seconds.

Solution:

Minutes	Seconds
1 5	3 5
+ 3 0	2 0
4 5	5 5

Therefore, sum is 45 min 55 sec.

Example 2: Subtract 25 minutes 32 seconds from 46 minutes 48 seconds.

Solution:

Minutes	Seconds
4 6	4 8
– 2 5	3 2
2 1	1 6

Therefore, difference is 21 min 16 sec.

EXERCISE 5.9

(1) Add:

i. 45 minutes 38 seconds and 30 minutes 40 seconds.
ii. 48 minutes 39 seconds and 37 minutes 20 seconds.
iii. 28 hours 10 minutes and 31 hours 20 minutes.
iv. 25 hours 20 minutes and 34 hours 15 minutes.

(2) Subtract:

i. 48 minutes 39 seconds from 59 minutes 49 seconds.
ii. 35 minutes 25 seconds from 55 minutes 35 seconds.
iii. 32 hours 12 minutes from 45 hours 46 minutes.
iv. 23 hours 10 minutes from 54 hours 30 minutes.

Solve simple real life problems involving conversion, addition and subtraction of units of time

Example 1: Nazia takes 1 hour 15 minutes to complete her Maths home work and 1 hour to complete her English home work. How much time she takes to complete both home works?

Solution:

	Hours	Minutes
Time taken to complete Maths homework:	1	15
Time taken to complete English homework:	+ 1	00
Total time taken:	**2**	**15**

Total time taken = 2 hours 15 minutes.

Example 2: Hamdan takes 1 hour 30 minutes to play cricket, while his brother Hammad takes 1 hour 15 minutes to play cricket. How much more time Hamdan takes to play cricket than his brother Hammad?

Solution:

	Hours	Minutes
Time taken by Hamdan	1	$\overset{2\,10}{\cancel{3}0}$
Time taken by Hammad	− 1	15
Difference	**0**	**15**

Hamdan takes 15 minutes more to play cricket.

EXERCISE 5.10

1. Rafique took 25 minutes 30 seconds to reach his school while 23 minutes 25 seconds to come back from school. How much total time he takes to go and come back from school?
2. Pakistani cricket team took 4 hours 25 minutes to complete their innings, while Indian cricket team took 3 hours 20 minutes to complete their innings. How much total time both the teams took to complete their innings?
3. Ahsan takes 42 minutes 54 seconds to complete a job, while his friend takes 32 minutes 12 seconds to complete the same job. How much more time does Ahsan take to complete the job?
4. A train takes 19 hours 48 minutes to reach from Lahore to Karachi, while another train takes 17 hours 23 minutes. Find the difference between the times taken by the two trains.
5. Momal spent 53 minutes 47 seconds to watch television while Zahid spent 39 minutes 23 seconds to watch the television. Find the difference in their time.

REVIEW EXERCISE – 5

1 **Tick (✓) the correct answer.**

(i) 1 centimetre is equal to
(a) 100 mm (b) 10 mm (c) 1000 mm

(ii) 1 kilometre is equal to
(a) 1000 m (b) 100 m (c) 10 m

(iii) 1 litre is equal to
(a) 100 *ml* (b) 10 *ml* (c) 1000 *ml*

(iv) The number of hours in a day is
(a) 12 (b) 24 (c) 30

2 **List any four lengths from your surrounding that would be measured by using km.**

Example:

(i) **Distance between my home and school is 2 km** (ii) ____________

(iii) ____________ (iv) ____________

3 **Convert these lengths into kilometres.**

(i) 6000 m = ________ (ii) 3500 m = ________

4 **Convert these lengths into metres.**

(i) 15 km = ________ (ii) 3 km = ________

5 **Add:**

(i) 30 km 43 m and 18 km 84 m
(ii) 48 m 65 cm and 38 m 76 cm
(iii) 13 *l* 800 *ml* and 12 *l* 700 *ml*
(iv) 44 kg 380 g and 38 kg 960 g

6 **Subtract:**

(i) 40 km 65 m from 76 km 72 m
(ii) 43 m 81 cm from 72 m 34 cm
(iii) 4 *l* from 12 *l* 250 *ml*
(iv) 45 kg 325 g from 86 kg 638 g

Unit 6

GEOMETRY

6.1 GEOMETRY BOX

Geometry Box contains different types of instruments which are used for measurement and drawing geometrical figures.

Know instruments of a Geometry Box. i.e., pencil, straight edge/ ruler, compasses, dividers, set squares and protractor.

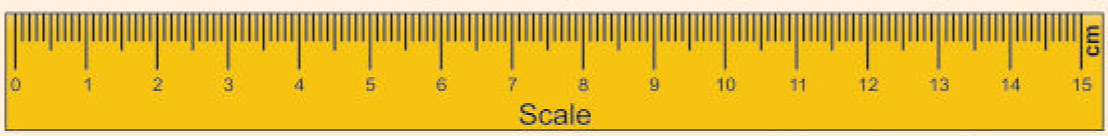
A geometry box contains the following instruments:

(1) Pencil and Eraser

Pencil is used for drawing figures and other lines, arcs, angles etc. and eraser is used to delete or correct the figures.

(2) Ruler (straight edge)

It is used to measure the length of a given line segment. It is also used in drawing line segment of given or required length.

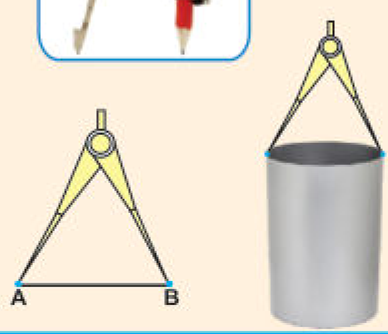


(3) Compasses (a pair of compasses)

Compasses are used to draw arcs, circles and marking distance.

(4) Dividers (a pair of dividers)

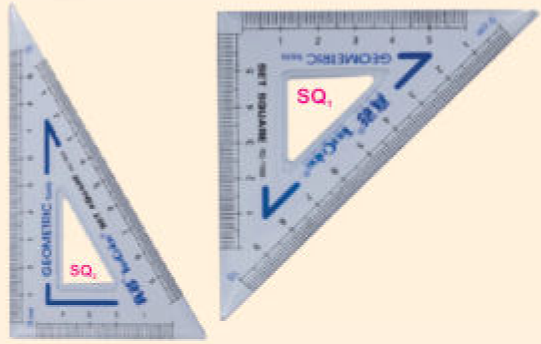
A pair of dividers is used to measure the length of a line segment and the diameter of hollow cylinder.



Teacher's Note Teacher should show the geometry box and instruments to the students in the class room.

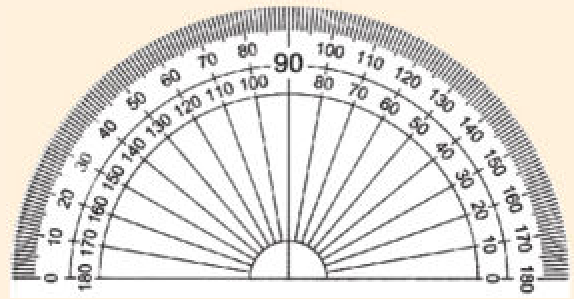
(5) Set squares

Set squares are used to draw a line parallel or perpendicular to a given line. It is also used to draw angles of 30°, 45°, 60° and 90°.



(6) Protractor

Protractor is used to measure an angle or to draw an angle of a given measure.



Recognize the use of pencils of grade H and HB

Pencils are characterized in two grades.

(i) Pencil of grade H
H stands for hardness

H grade pencil

The line drawn with pencil H grade is very thin. In this pencil the lead leaves light black colour impression.

(ii) Pencil of grade HB.
HB stands for blackness

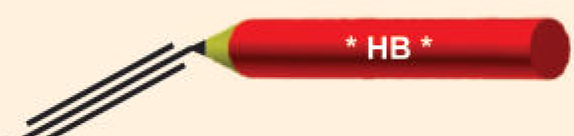
HB grade pencil

The line drawn with HB grade is bold. In this pencil the lead leaves dark black colour impression.

Demonstrate the use of H and HB pencils by drawing different lines

Look at these pencils

The line drawn with pencil of grade H is very thin.



The line drawn with pencil of grade HB is bold.

Activity 1 **Draw the following diagrams with *H* and *HB* grade pencils.**

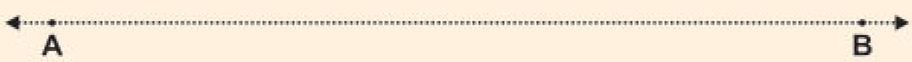
(i) Line segment **(ii)** Square **(iii)** Circle

Activity 2 Take a point A. From point A draw $\overline{AB}$ with the help of any straight edge. Draw another $\overline{AC}$ from point A. How many lines can be drawn from point?

Activity 3 Take a pair of dividers. Penetrate the pointed ends on a paper. Name the two marks as A and B. Now draw lines with the help of straightedge; so that they may pass through both the points A and B.

6.2 LINE

In previous class, we learnt that a line consists of a set of infinite points. A line has no end point.

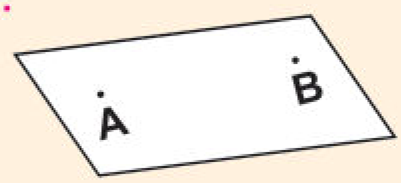


It is a line which shows infinite number of points.

Measure the length of a line in centimetres and millimetres using straightedge/ruler and dividers

Activity 1 **To draw a line segment using straightedge / ruler.**

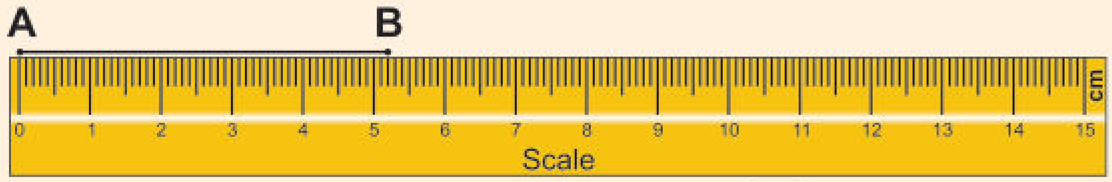
Step 1. Take two points (say A and B).
Step 2. Join A and B, using your ruler and pencil.



Step 3. Thus we get a part of a line which has two end points called the line segment.

Teacher's Note Teacher may organise activities and engage the students to draw and measure line segment in their own copies.

Activity 2 Measuring the length of a line segment with ruler.

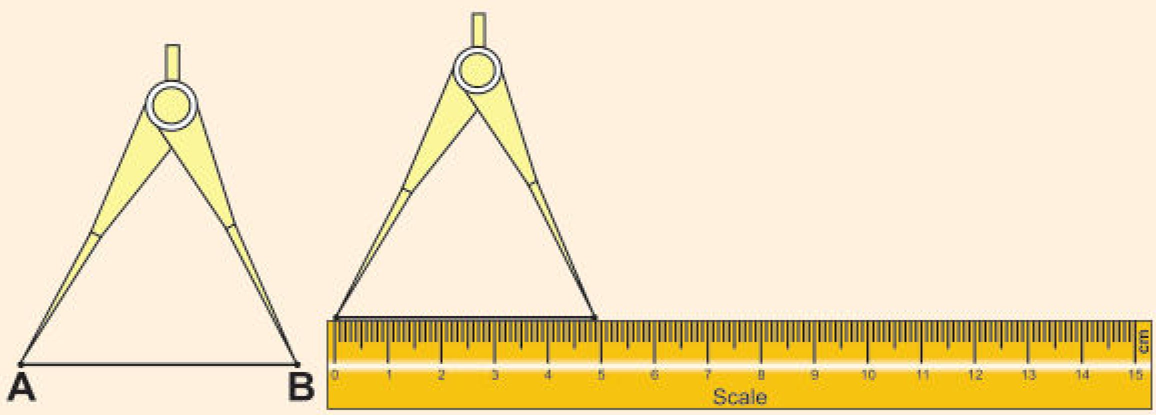


Step 1. Place a ruler with its edge along $\overline{AB}$ such that zero (0) mark of the ruler faces the point A.

Step 2. Read the mark on the ruler which faces the point B.

Step 3. This gives us the length of AB. Thus the length of $\overline{AB}$ is 5 cm 2 mm i.e. 5.2 cm. Symbolically, we write m $\overline{AB}$ = 5.2 cm.

Activity 3 Measuring the length of line segment with divider.



Step 1. Open the divider so that the end point of one of its arms is at A and the point of the second arm is at B.

Step 2. Lift the divider without disturbing it and place it on the ruler so that the end point of one arm is at zero (0) mark.

Step 3. Read the mark against the end point of the second arm of the divider.

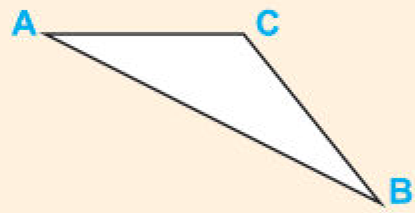
Step 4. We find the length of $\overline{AB}$ to be 4.9 cm or we write m $\overline{AB}$ = 4.9 cm

Unit 6 GEOMETRY (Line)

EXERCISE 6.1

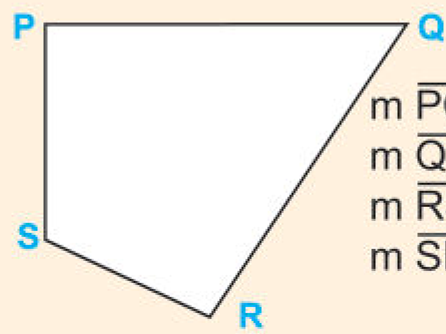
1. Measure the sides of the following figures with ruler and write their lengths:

(i)



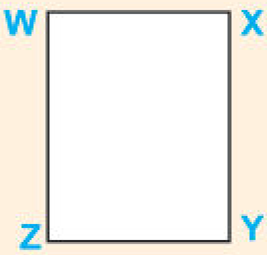
m $\overline{AC}$ = ________cm
m $\overline{AB}$ = ________cm
m $\overline{CB}$ = ________cm
m $\overline{BC}$ = ________cm

(ii)



m $\overline{PQ}$ = ________cm
m $\overline{QR}$ = ________cm
m $\overline{RS}$ = ________cm
m $\overline{SP}$ = ________cm

(iii)



m $\overline{WX}$ = ___ = m $\overline{YZ}$ = ___
m $\overline{XY}$ = ___ = m $\overline{ZW}$ = ___

2. Join the pair of points given below, to draw line segments, then measure the length of each of the line segments with the ruler and write its length. Verify the length by divider.

(i) P ________________________ Q
Length of $\overline{PQ}$ is 8.5 cm

(ii) S . . T
Length of $\overline{ST}$ is_____cm

(iii) F . . G
Length of $\overline{FG}$ is_____cm

(iv) Y . . Z
Length of $\overline{YZ}$ is_____cm

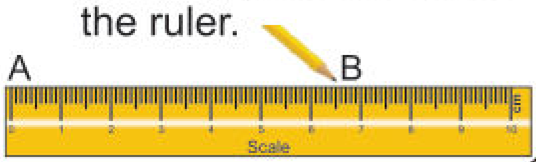
(v) M . . N
Length of $\overline{MN}$ is_____cm

Draw a straight line of given length using a straightedge/ruler and dividers

Activity 1 **Let's draw a line segment 6.5 cm long using ruler.**

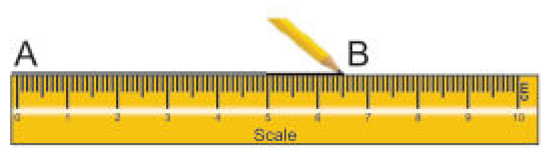
1st Step

(i) Take any point A.
(ii) Place the 0 (zero) of the ruler against point A.
(iii) Put another point B with pencil against 6.5 cm of the ruler.



2nd Step

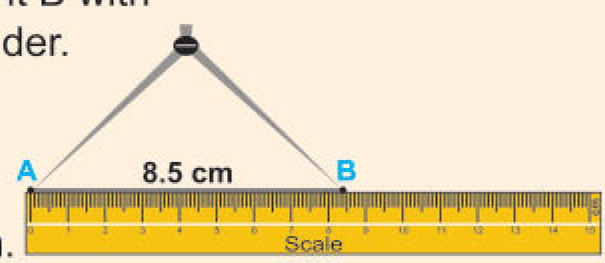
(i) Join points A and B using the ruler or any straight edge.



Thus a straight line AB is drawn whose length is 6.5 cm

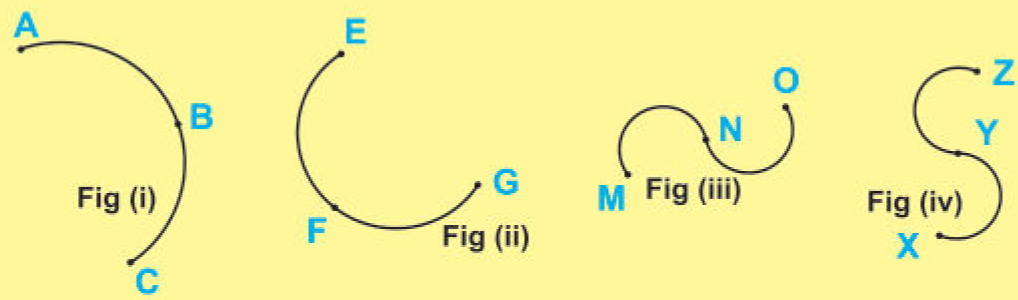
Activity 2 **To draw a straight line of 8.5 cm using dividers.**

Step 1. Take a point A on a sheet of paper.
Step 2. Place one end of the divider at zero mark on ruler.
Step 3. Open the divider so that the other end of the divider is on the mark of 8.5 cm on the ruler.
Step 4. Without changing the openings of dividers, place one end at A and put a point B with another end of the divider.
Step 5. Join A and B.
Step 6. Thus, we obtain a straight line AB of required length 8.5 cm.



Teacher's Note Help the students to draw lines of different measure in their own copy.

Draw a curved line and measure its length using thread/dividers and straightedge / ruler.



These are curved lines. Curved lines can be drawn by moving pencil in different directions. We can measure its length by using thread/divider and ruler. Measure other curved lines.

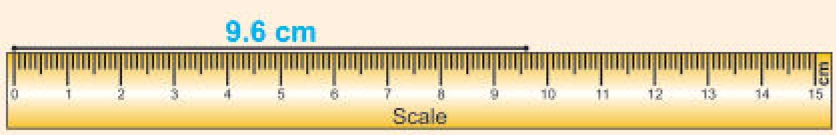
Activity 1 Measure curved line of Fig (i) with thread and ruler.

Step 1. Take a piece of thread.

Step 2. Place one end of the thread at point A.

Step 3. Spread the thread along the path from A to B and then from B to C.

Step 4. Put a mark on the thread or cut it with a pair of scissors.

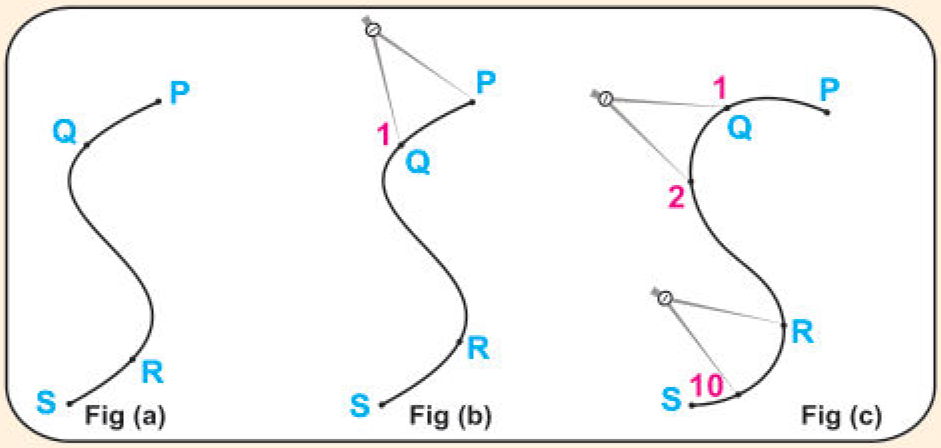


Step 5. Measure the length of the thread with the help of ruler which is **9 cm 6 mm** or **9.6 cm**.

Step 6. In this way the length of the said curved line **ABC** is **9.6 cm**.

Teacher's Note Teacher may help the students to draw some curved lines in their copies and measure their lengths with thread or scale.

Activity 2 **Measure curved line PQRS fig. (a) with dividers and ruler.**



Step 1: Open the arms of divider **1 cm** apart.

Step 2: Place one arm at **P** so that the other arm reaches at **1** see fig (b).

Step 3: Hold firm arm at No. **1**, rotate the line arm to fall at No. **2** see fig (c).

Step 4: Repeat the above process again and again to reach at No. **10** see fig. (c).

Step 5: Remaining part of the curved line is not a complete unit. Open divider touching points S and No. **10**. Place it on the ruler so that one arm is at **0** (zero). Read the other point. Suppose the second arm falls at **5 mm**. Hence the required length of the curved line **PQRS** is **10.5 cm.**

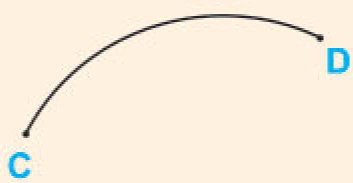
EXERCISE 6.2

1. Draw line segments of following lengths. Using (a) ruler (b) ruler and dividers.

(i) $m\overline{AB} = 7.4$ cm	(ii) $m\overline{BC} = 6.6$ cm	(iii) $m\overline{CD} = 5.7$ cm
(iv) $m\overline{DE} = 3.8$ cm	(v) $m\overline{EF} = 4.9$ cm	(vi) $m\overline{PQ} = 6.0$ cm

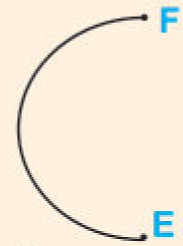
2. Measure these curved lines with thread, ruler and dividers and write their lengths.

(i)



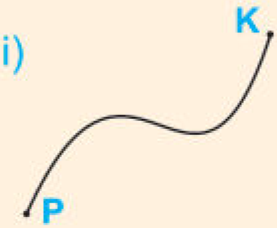
Length _____ cm

(ii)



Length _____ cm

(iii)



Length _____ cm

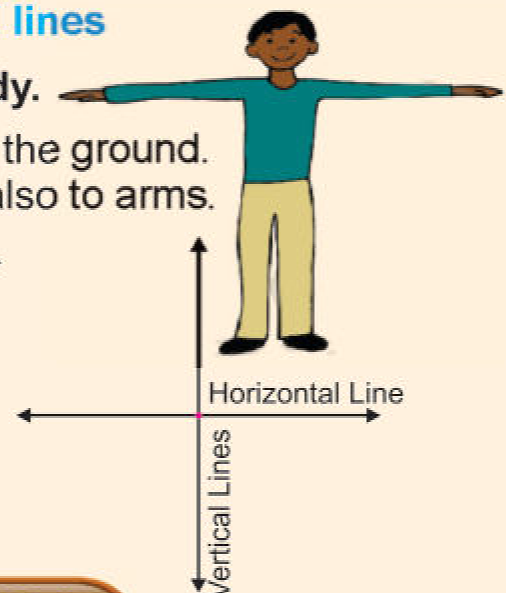
Recognize horizontal and vertical lines

Look at stretched arms of the body.

Arms represent a line **horizontal** to the ground. Body is **vertical** to the ground and also to arms.

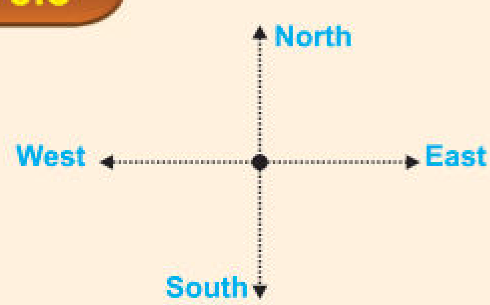
Thus we get a horizontal line ⟷ and a vertical line ↕ which intersect each other at a point.

Note: Horizontal and vertical lines have arrow marks, which represent direction.



EXERCISE 6.3

Look at the directions North, South, East and West shown on Horizontal and Vertical lines and fill in the blanks.



(1) Horizontal line shows____________directions.

(2) Vertical line shows____________directions.

(3) North to South direction represents____________line.

(4) West to East direction represents____________line.

Teacher's Note Teacher may show the students horizontal and vertical lines with back of chair, window pan, sides of table, blackboard and corner of book with both hands of a clock at the time 6 o'clock and 15 minutes past.

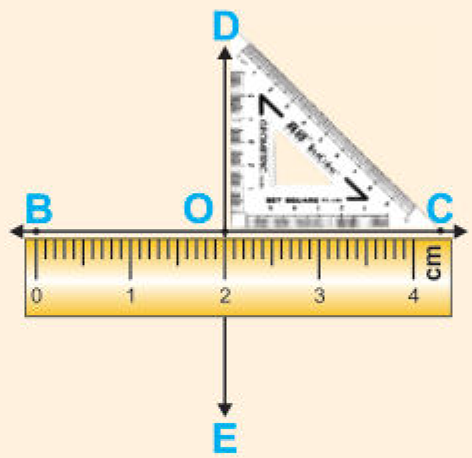
Draw a vertical line on a given horizontal line using set squares

Activity **Draw a vertical line $\overleftrightarrow{DE}$ on a given horizontal line $\overleftrightarrow{BC}$.**

Step 1. Draw a horizontal line $\overleftrightarrow{BC}$.

Step 2. Place ruler edge along $\overleftrightarrow{BC}$.

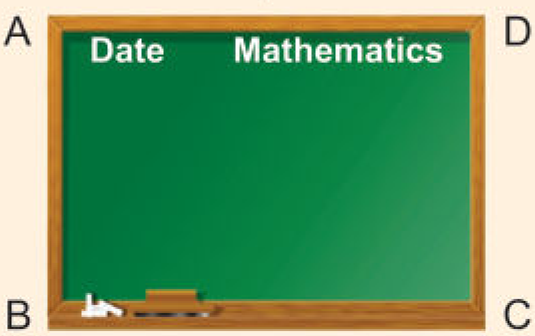
Step 3. Along the side of the ruler edge, place a set square. Now slide it until its square corner meet the point O at which the vertical line is to be drawn.



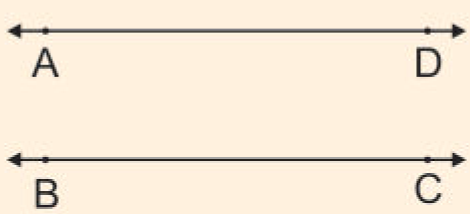
Step 4. Draw $\overleftrightarrow{DO}$ as shown in the figure. Then extend $\overrightarrow{DO}$ to E. Thus we get a vertical line $\overleftrightarrow{DE}$ on a given horizontal line $\overleftrightarrow{BC}$.

Recognize parallel and non-parallel lines

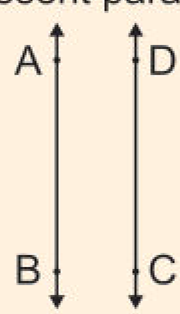
Look at these pictures:



The opposite edges of the black board represent parallel lines;

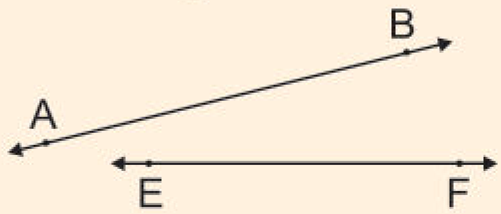


and

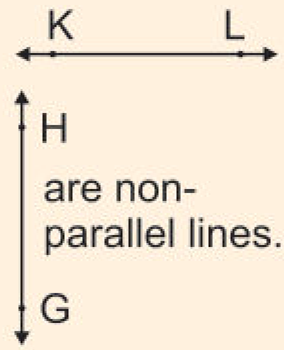


$\overleftrightarrow{AD}$ is parallel to $\overleftrightarrow{BC}$ and $\overleftrightarrow{AB}$ is parallel to $\overleftrightarrow{DC}$. Similarly the two lines of the railway track are parallel.

On the contrary, the following pair of lines are non-parallel lines because they will meet or intersect if extended.



and

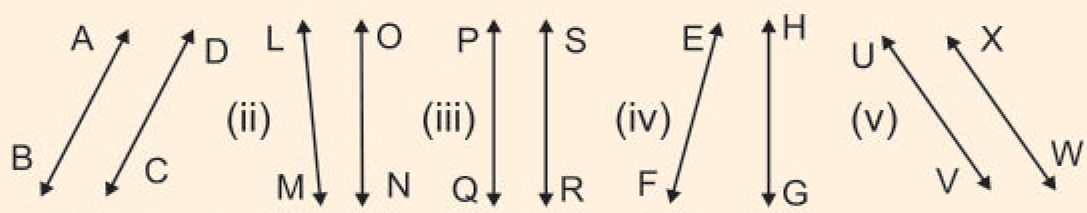


are non-parallel lines.

Hence parallel lines are those lines which do not intersect, however long they are extended.

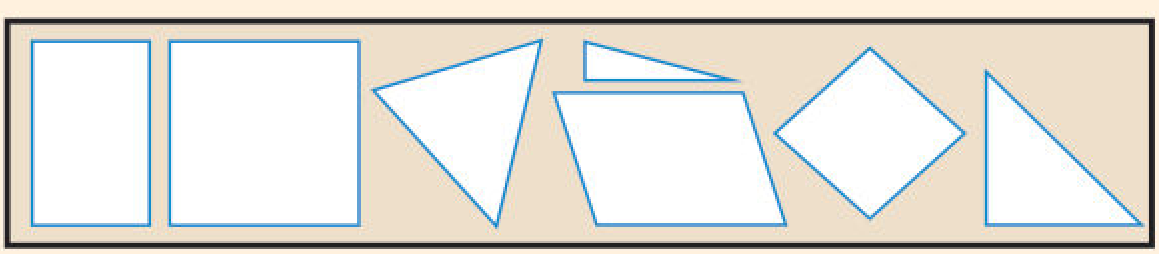
Identify parallel and non-parellel lines from a given set of lines

Example: Identify parallel and non-parallel lines.



Here (i), (iii) and (v) are all pair of parallel lines.
But (ii) and (iv) are pair of non-parallel lines.

Cross (✗) the shapes which represent parallel lines and (✓) the shapes which represent non-parallel lines.



Teacher's Note

Teacher should give examples from real life situation to recognized the parallel and non-parallel lines.

Activity 2 **Write down some pairs of:**

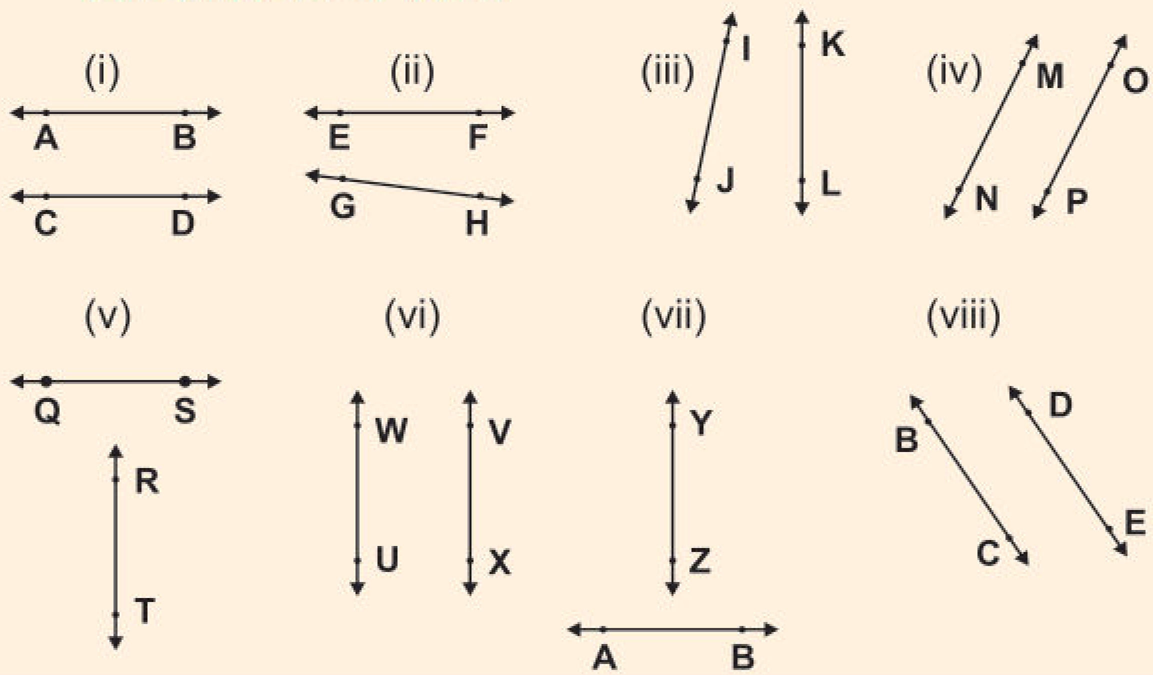
Parallel lines	Non-Parallel lines
1. Edges of the black board	1. Sides of triangle
2.	2.
3.	3.
4.	4.
5.	5.

Observations:

(1) Pair of parallel lines never meet; how far they are extended.
(2) Pair of non-parallel lines will meet and intersect each other.

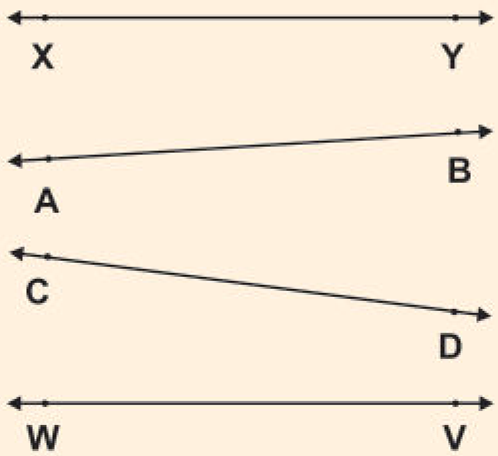
EXERCISE 6.4

1. Identify parallel and non-parallel lines from the following set of lines.

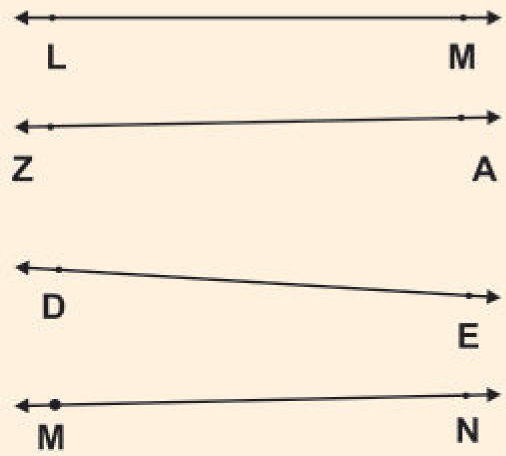


Teacher's Note Teacher may ask the students to verify the above mentioned observations in their copies.

2. Tick (✓) the lines which are parallel to $\overleftrightarrow{XY}$?



3. Cross (✗) the lines which are non-parallel to $\overleftrightarrow{LM}$?

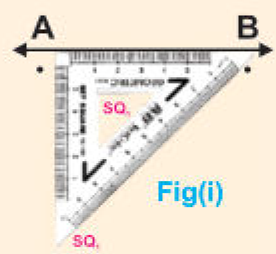


Draw a parallel line to a given straight line using set squares

Activity Draw a line (or lines) parallel to $\overleftrightarrow{AB}$.

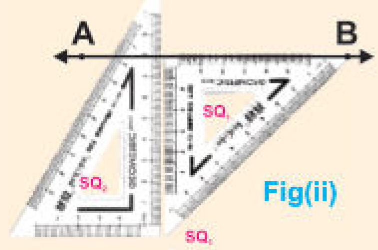
Step 1. Draw given lines $\overleftrightarrow{AB}$.

Step 2. Place the edge of one set square (SQ_1) along $\overleftrightarrow{AB}$ as shown in fig. (i)



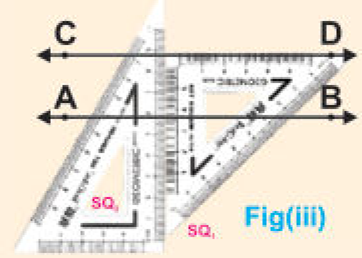
Fig(i)

Step 3. Place another set square (SQ_2) adjacent to the previous one (fig ii). Now SQ_1 is ready to slide up and down along SQ_2



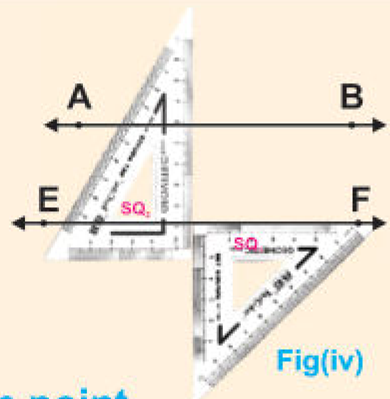
Fig(ii)

Step 4. Holding firm SQ_2 slide up SQ_1 and draw $\overleftrightarrow{CD}$ as shown in fig (iii) Therefore $\overleftrightarrow{AB}$ is parallel to $\overleftrightarrow{CD}$.



Fig(iii)

Step 5. Holding firm SQ_2 slide down SQ_1 and draw $\overleftrightarrow{EF}$ as shown in fig (iv)
Therefore $\overleftrightarrow{AB}$ is parallel to line $\overleftrightarrow{EF}$.



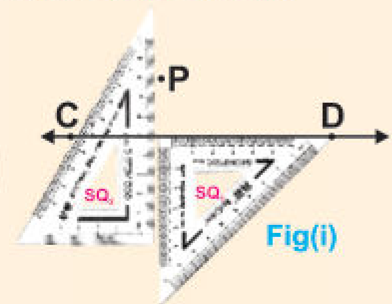
Fig(iv)

Draw a line which passes through a given point and isparallel to a given line (using set - squares)

Activity **Draw a line $\overleftrightarrow{RS}$ parallel to a given line $\overleftrightarrow{CD}$ and passing through a given point P.**

Step 1. Draw given $\overleftrightarrow{CD}$ and take point P out side it.

Step 2. Place the set squares as explained previously and shown in figure (i).

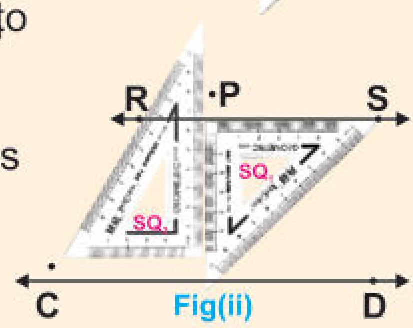


Fig(i)

Step 3. Holding fast SQ_2 slide up SQ_1 to reach at point P.

Step 4. Draw $\overleftrightarrow{RS}$ passing through P as shown in figure (ii).

Therefore $\overleftrightarrow{CD}$ is parallel to $\overleftrightarrow{RS}$ which is passing through the point P.



Fig(ii)

EXERCISE 6.5

1. Draw a vertical line $\overleftrightarrow{PQ}$ on a given horizontal line $\overleftrightarrow{XY}$; using set squares.
2. Draw a parallel line $\overleftrightarrow{YZ}$ to a given line $\overleftrightarrow{PQ}$ using set squares.
3. Draw $\overleftrightarrow{AB}$ which passes through a given point E and is parallel to a given $\overleftrightarrow{CD}$ (using set squares)

6.3 ANGLE

Recognize an angle through non-parallel lines

Look at the two non-parallel lines $\overleftrightarrow{AE}$ and $\overleftrightarrow{CD}$ fig (i)

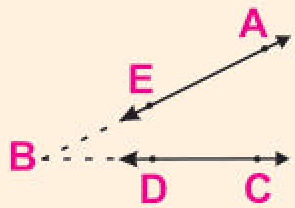
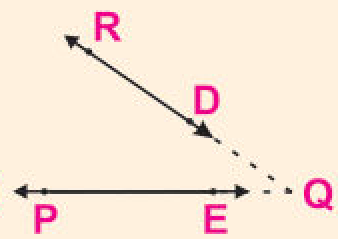


Fig (i)

Fig (ii)



These lines are produced to meet at point B and make an angle ABC. Thus two non-parallel lines have a common end point. Here the common end point is B.

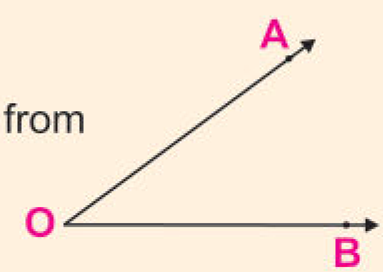
Again look at the two non-parallel lines $\overleftrightarrow{PE}$ and $\overleftrightarrow{RD}$ fig (ii). These lines are produced to meet at point Q and make an angle PQR. Here Q is the common end point of line $\overleftrightarrow{RQ}$ and $\overleftrightarrow{PQ}$

Draw an angle AOB with vertex (O) and arms ($\overrightarrow{OA}$, $\overrightarrow{OB}$) to recognize the notation ∠AOB for an angle AOB

Activity **Draw an angle.**

Step 1: Draw $\overrightarrow{OB}$.

Step 2: Draw another $\overrightarrow{OA}$ (not along $\overrightarrow{OB}$) from point O. This is an angle AOB (or angle BOA)



The common end point O is the vertex of angle AOB.

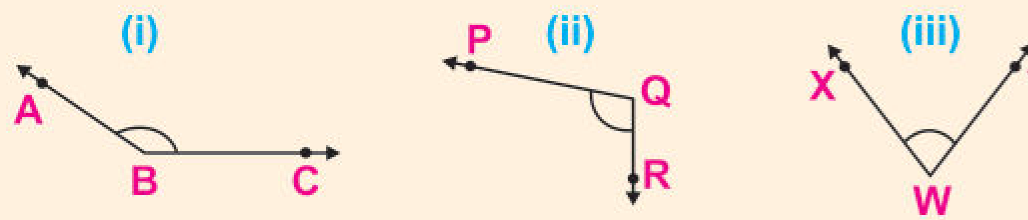
$\overrightarrow{OA}$ and $\overrightarrow{OB}$ are arms of angle AOB.

The symbol for angle is∠

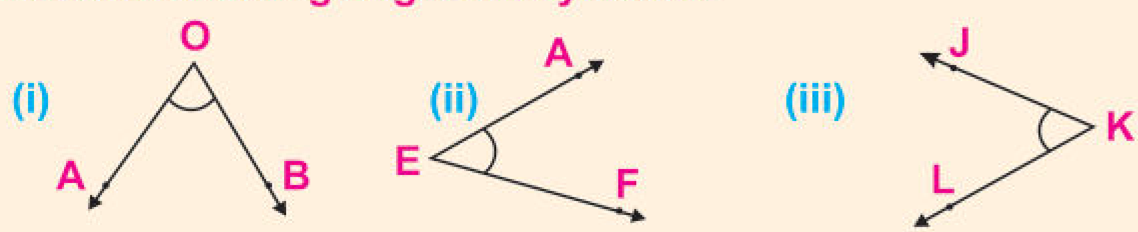
So angle AOB is written as ∠AOB or ∠BOA

EXERCISE 6.6

1. Write the names of vertex and arms each of the following angles.



2. Write the following angles in symbols:



Recognize right angle through horizontal and vertical lines

Look at the following pairs of horizontal and vertical lines.

When vertical and horizontal lines meet at a point they form a right angle. In figure (i) ABC is a right angle.

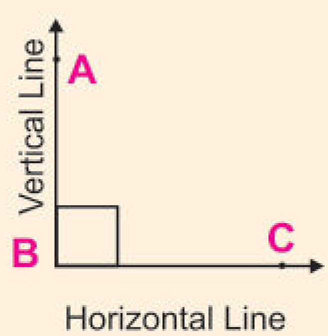


Fig (i)

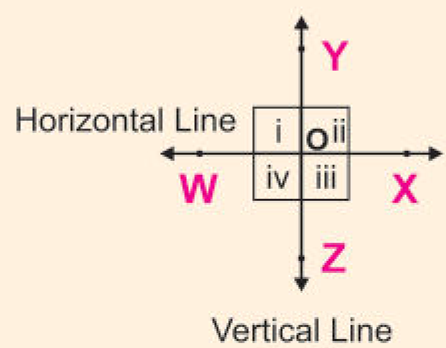


Fig (ii)

In figure (ii), the pairs of horizontal and vertical lines intersect each other at a point O and form four right angles. Hence

(i) $\angle$WOY is a right $\angle$ (ii) $\angle$XOY is a right $\angle$

(iii) $\angle$XOZ is a right $\angle$ (iv) $\angle$ZOW is a right $\angle$

Activity 1 To make four right angles by folding a paper sheet.

Step 1. Take a piece of paper. Fold it into two halves and then into four quarters.

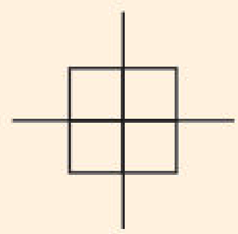
Step 2. Draw lines on the creases of the paper.

Step 3. Name horizontal line as $\overleftrightarrow{CD}$ and vertical line as $\overleftrightarrow{AB}$. They intersect each other at point O.

Step 4. They form four right angles.

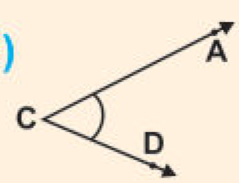
Step 5. We can write their names symbolically.
$\angle AOC$, $\angle COB$, $\angle BOD$ and $\angle DOA$.

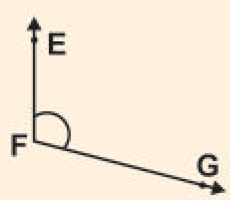
Note: We can draw square in each right angle at its vertex.

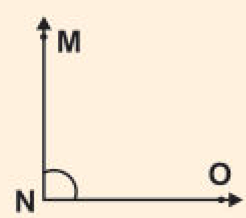


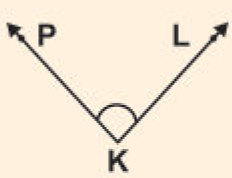
EXERCISE 6.7

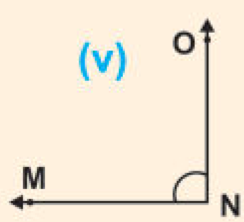
1. Look at the following angles and tick (✓) all those that are right angles.

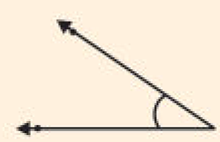
(i)


(ii)


(iii)


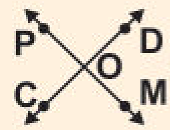
(iv)


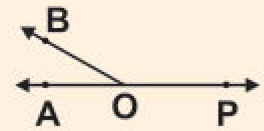
(v)


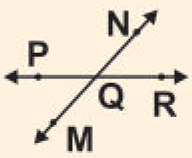
(vi)


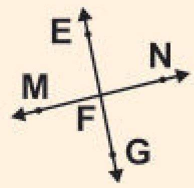
2. Which of the following figures show right angles?

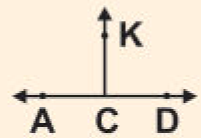
(i) 

(ii) 

(iii) 

(iv) 

(v) 

(vi) 

Demonstrate acute and obtuse angles via the right angle

Activity 1 **Draw a right angle.**

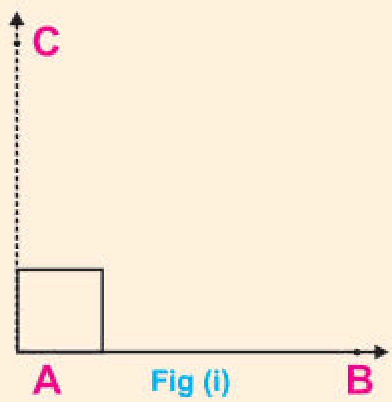


Fig (i)

Step 1. Draw a horizontal line $\overrightarrow{AB}$.

Step 2. At point A, draw a dotted vertical line $\overrightarrow{AC}$.

Step 3. So $\angle BAC$ is a right angle. See fig (i)

Activity 2 **Draw an acute angle.**

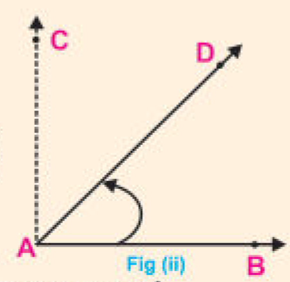


Fig (ii)

Step 1. Draw a third line $\overrightarrow{AD}$ between $\overrightarrow{AC}$ and $\overrightarrow{AB}$ as shown in fig (ii).

Step 2. We have another angle BAD (or $\angle$ DAB).

Step 3. $\angle$BAD is smaller than $\angle$BAC because the curved arrow ( ) is stopped by arm $\overrightarrow{AD}$ before reaching the arm $\overrightarrow{AC}$.

Step 4. Thus $\angle$BAD is less than a right angle $\angle$BAC.

Step 5. Hence $\angle$BAD is the required acute angle.

> An angle which is less than a right angle is called an acute angle.

Activity 3 Draw an obtuse angle.

Step 1. Draw a line $\overrightarrow{AF}$ outside the right angle BAC to get another angle BAF, as shown in figure (iii).

Step 2. ∠BAF is greater than ∠BAC because the curved arrow goes beyond arm $\overrightarrow{AC}$ to reach the arm $\overrightarrow{AF}$.

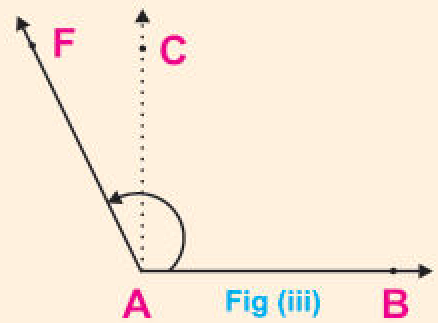


Fig (iii)

Step 3. Thus ∠BAF is greater than a right angle.

Step 4. Hence ∠BAF is an obtuse angle.

An angle which is greater than 90º but less than 180º, is called an obtuse angle.

EXERCISE 6.8

1. Look at the previous figures (i), (ii) and (iii) and complete the following sentences.

(i) ∠BAC is ______ angle. (ii) ∠BAD is ______ angle.
(iii) ∠BAF is ______ angle. (iv) ∠DAB is ______ angle.
(v) ∠FAB is ______ angle. (vi) ∠CAB is ______ angle.

2. Draw the following angles.

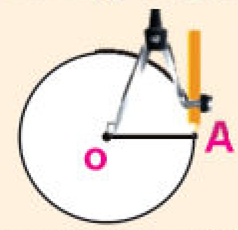
(i) ∠ABC (obtuse angle)
(ii) ∠PQR (acute angle) (iii) ∠ABC (right angle)
(iv) ∠DEF (obtuse angle) (v) ∠WXY (right angle)

Unit 6 GEOMETRY (Angle)

Recognize the standard unit for measuring angles as one degree (1°) which is defined as $\frac{1}{360}$ of a complete revolution

Activity **Define standard unit for measuring angle.**

Step 1. Take O as centre and radius $\overline{OA}$, with a pair of compasses. Draw a complete revolution, it describes a circle.



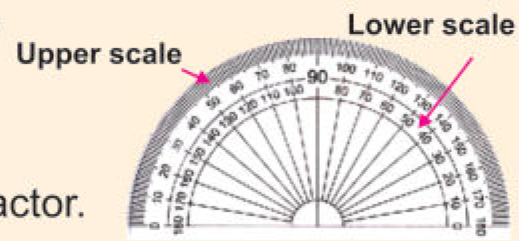
Step 2. Divide this circle in 360 equal parts.
Each equal part is called a degree. It is denoted by "1°".

Step 3. The number of degrees in a complete turn are 360°.

Step 4. A "degree" is $\frac{1}{360}$ th part of a complete revolution.

Measure angles using protractor

Protractor is used to measure angle from 0° to 180°. There are two scales of number marked on protractor.



The upper scale of protractor reads the measure of angle from left to right. The lower scale of protractor reads the measure of angle from right to left.

Activity 1 **Measure the given acute angle ∠PQR**

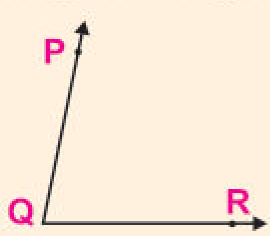
Step 1. Put the centre of the protractor on the vertex Q of ∠PQR.

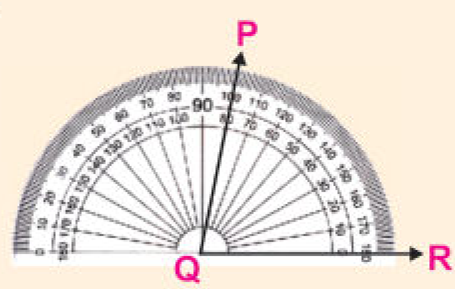
Step 2. Base line coincides with one arm $\overrightarrow{QR}$.

Step 3. Start reading from 0° in lower scale from right to left.

Step 4. Mark at a point on which arm $\overrightarrow{QP}$ lies.

Step 5. Thus m∠PQR = 80°





Activity 2 **Measure the given obtuse angle ∠DEF.**

Step 1. Put the centre of the protractor on the vertex E of ∠DEF.

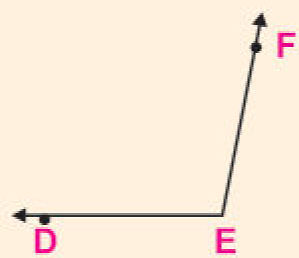
Step 2. The base line coincides with one arm $\overrightarrow{ED}$.

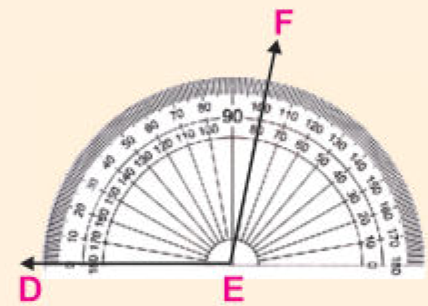
Step 3. Start reading from 0 in upper row from left to right.

Step 4. Mark at a point on which arm $\overrightarrow{EF}$ lies.

Step 5. The point F crosses the number 100.

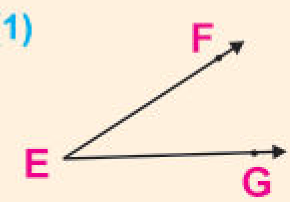
Thus m∠DEF = 100°.

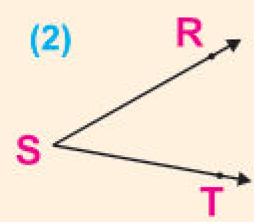


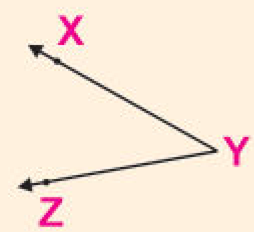


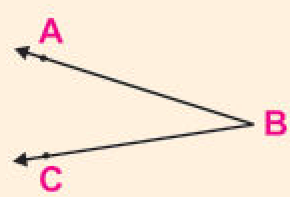
EXERCISE 6.9

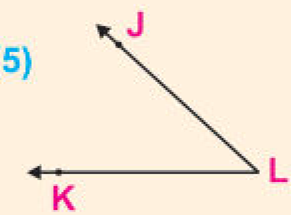
Using the protractor, measure the following angles.

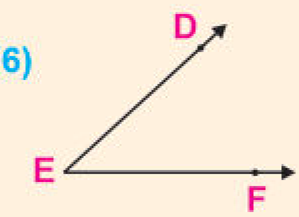
(1)


(2)


(3)


(4)


(5)


(6)


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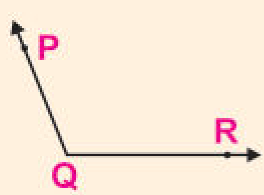

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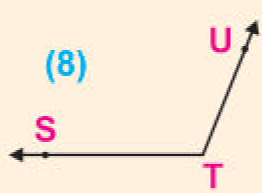

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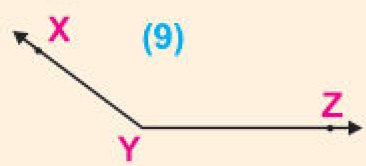


Unit 6 GEOMETRY (Angle)

Draw a right angle using protractor

Activity **Draw an angle BAC of 90°.**

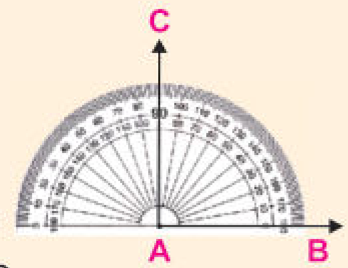


Fig (i)

Step 1. Draw $\overrightarrow{AB}$ horizontally.

Step 2. Place protractor on $\overrightarrow{AB}$ such that the middle of its bottom line is exactly on A.

Step 3. Find the 90° mark on the protractor. Take a point against it, and name it C, fig (i).

Step 4. Join C to A.

Step 5. We get ∠BAC as shown in fig (ii).

Step 6. It is the required angle of 90° known as right angle.

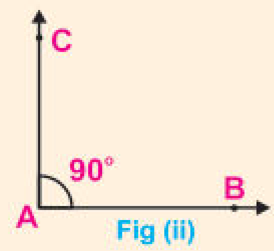
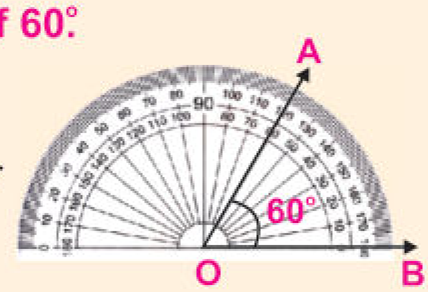


Fig (ii)

Draw acute and obtuse angles of different measures using protractor

Activity 1 **Draw an acute angle of 60°.**



Step 1. Draw $\overrightarrow{OB}$.

Step 2. Place the centre of the protractor at point O, one end of $\overrightarrow{OB}$.

Step 3. Adjust the protractor so that the line of the 0 (zero) mark on the right side coincides with $\overrightarrow{OB}$.

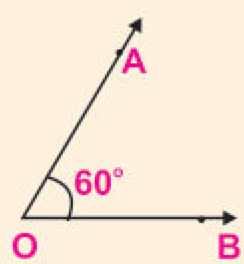
Step 4. Extend $\overrightarrow{OB}$ if necessary.

Step 5. Make a fine point mark against the 60° mark on the protractor.

Step 6. Name this point A.

Step 7. Draw $\overrightarrow{OA}$ and extend it.

Step 8. AOB is the required acute angle measuring 60°.



Activity 2 **Draw an obtuse angle of 130°.**

Step 1. Draw a $\overrightarrow{YZ}$.

Step 2. We have to draw an angle of 130° at point Y.

Step 3. Place the centre of the protractor on the point Y.

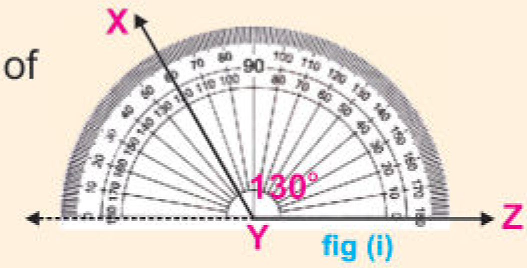


fig (i)

Step 4. Let 0 (zero) mark on the right side of the protractor be exactly on $\overrightarrow{YZ}$.(Extend $\overrightarrow{YZ}$ if necessary).

Step 5. Find 130° mark on the protractor. Take a point against it and call it X; see fig (i).

Step 6. Draw $\overrightarrow{YX}$ and extend it.

Thus ∠XYZ is the required angle of 130°, see fig (ii).

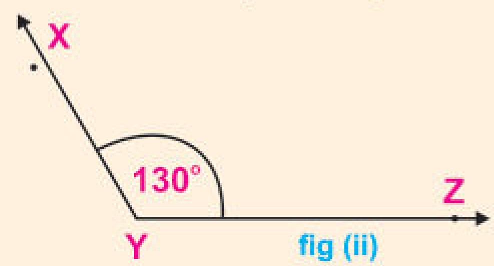
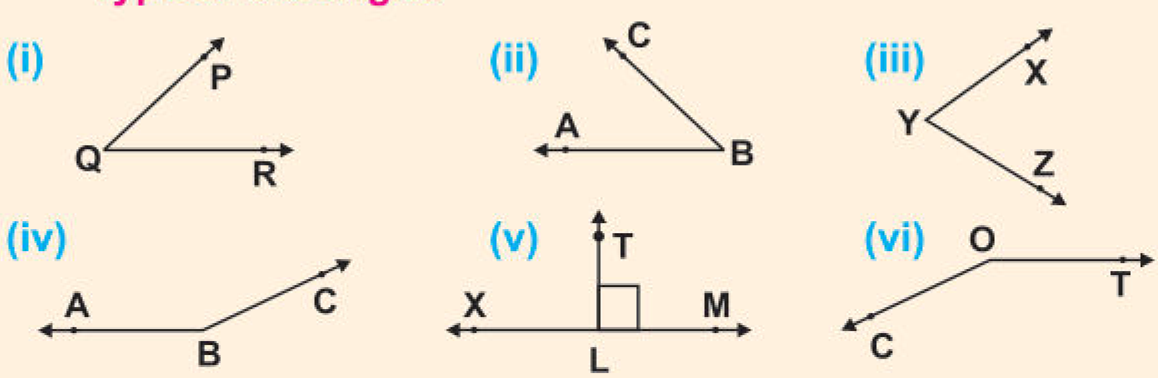


fig (ii)

EXERCISE 6.10

(1) Measure each of the following angles and then tell the type of the angle.



(2) Draw the following angles with the help of protractor.

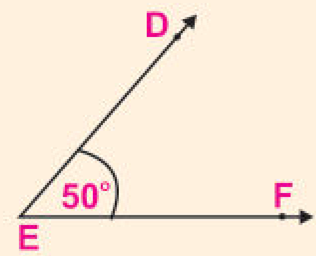
(i) 10°	(ii) 40°	(iii) 20°	(iv) 60°	(v) 30°
(vi) 80°	(vii) 90°	(viii) 120°	(ix) 145°	(x) 45°

Draw an angle (using protractor)

We have already learnt to draw an angle of any measure with the help of protractor. Also we have learnt to measure any given angle with the help of protractor.

(a) Draw an angle equal in measure to a given angle.

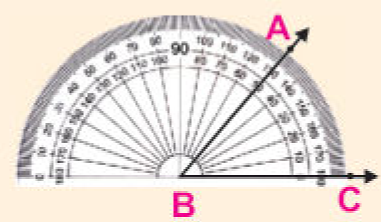
Steps of construction:



Step 1. Measure the given $\angle DEF$ with the help of protractor. It is found that $m\angle DEF = 50^\circ$.

Step 2. We have to draw another angle say $\angle ABC$ such that

$m\angle ABC = m\angle DEF = 50^\circ$.

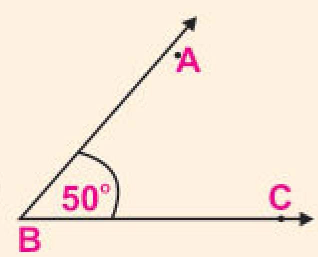


Step 3. Draw $\overrightarrow{BC}$ with initial point B.

Step 4. Place the centre of the protractor on B and adjust the base line of protractor to coincide with $\overrightarrow{BC}$.

Step 5. Start from zero and read the lower scale of the protractor up to 50°.

Step 6. Mark a point A against 50° mark.

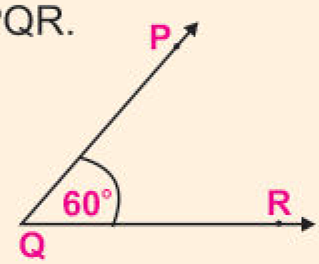


Step 7. Remove the protractor and draw $\overrightarrow{BA}$.

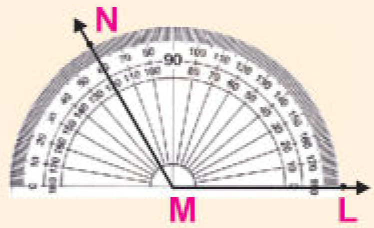
Thus $m\angle ABC = 50^\circ$. It is equal in measure to given $\angle DEF$.

(b) Draw an angle twice the measure of a given angle.

First of all we have to measure the given $\angle PQR$.
Let the measure of given angle is 60°.
Therefore we have to draw an angle of measure $2 \times 60^\circ = 120^\circ$
Say $m\angle LMN = 120^\circ$

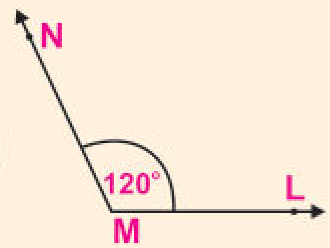


Steps of construction:



Step 1. Draw $\overrightarrow{ML}$ with M as initial point.

Step 2. Place the centre of the protractor on M and adjust the base line of protractor to coincide with $\overrightarrow{ML}$.

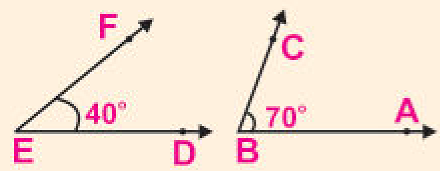


Step 3. Start from zero and read the lower scale on the protractor up to 120°.

Step 4. Mark a point N against 120° mark.

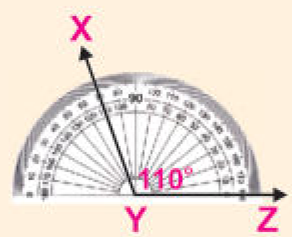
Step 5. Remove the protractor and draw $\overrightarrow{MN}$
Thus we get $\angle LMN$ such that
$m\angle LMN = 2 \times (m\angle PQR) = 120°$

(c) Draw an angle equal in measure the sum of two angles.



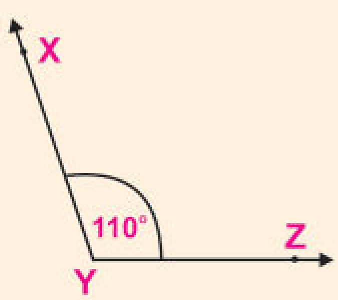
First of all we have to measure the given angles, $\angle ABC$ and $\angle DEF$ with the help of protractor.
Let $m\angle DEF = 40°$ and $m\angle ABC = 70°$
The sum of two given angles is $40° + 70° = 110°$
Now we have to draw $\angle XYZ$ such that $m\angle XYZ = 110°$

Steps of construction:



Step 1. Take an initial point Y and draw $\overrightarrow{YZ}$.

Step 2. Place the centre of the protractor on Y and adjust the base line of protractor to coincide with $\overrightarrow{YZ}$.



Step 3. Start from zero, read the lower scale of protractor up to 110°.

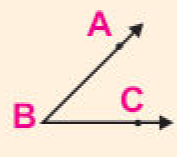
Step 4. Mark a point X against 110°.

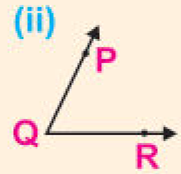
Step 5. Remove protractor and draw $\overrightarrow{YX}$.

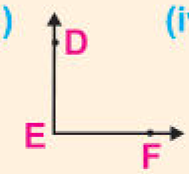
Thus we get $\angle XYZ$ such that
$m\angle XYZ = m\angle DEF + m\angle ABC = 110°$

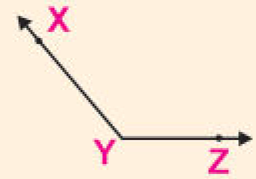
EXERCISE 6.11

(1) Draw angles with the help of protractor equal in measure to the given angle.

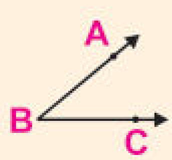
(i)


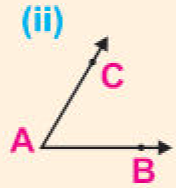
(ii)


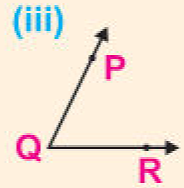
(iii)


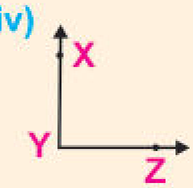
(iv)


(2) Draw angles with the help of protractor twice the measure of the given angle.

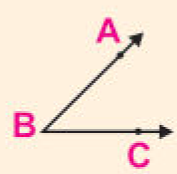
(i)


(ii)


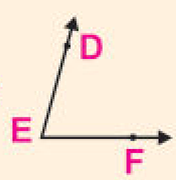
(iii)


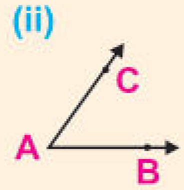
(iv)


(3) Draw angles equal in measure to the sum of two angles. (with the help of protractor).

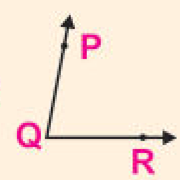
(i)


and



(ii)


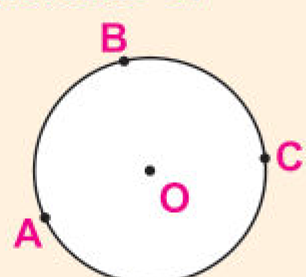
and



6.4 CIRCLE

Look at the picture. It is the picture of a circle. There are three points A, B and C on the circle. Its centre is O. Points A, B and C are at the same distance from O.

Identify centre, radius, diameter and circumference of a circle

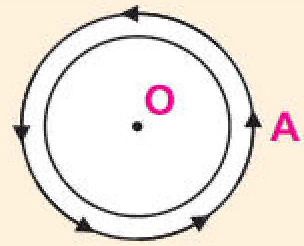


(i) Centre: All the points of a circle are at the same distance from a fixed point O, called its centre.

Example:

Point A, B and C are at the same distance from centre O.

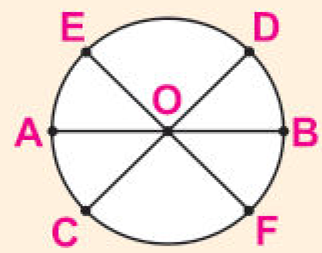
(ii) Circumference: The length of the circle is called the **circumference** of the circle. It is the distance that we cover by taking exactly one complete round of the circle.



Example:

Lets start at point A and again reach the same point A after completing one revolution. This distance is the circumference of circle.

(iii) Diameter: The line segment passing through the centre of the circle and joining the circle at two different points is called the **diameter**.



Example:

$\overline{AB}$, $\overline{CD}$, $\overline{EF}$ etc are the diameters.

(iv) Radius: Radius is half of the length of the diameter of a circle.

Remember: $\boxed{\text{Radius} = \dfrac{\text{length of the diameter}}{2}}$

Draw a circle of given radius using compasses and straightedge/ruler

Draw a circle using compass whose radius is 2 cm.

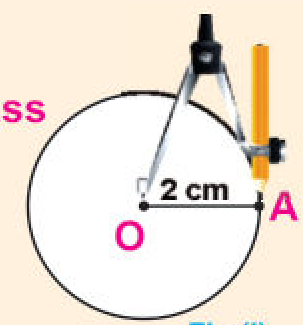


Fig (i)

Step 1. Draw $\overline{OA}$, 2 cm long.

Step 2. Take O as centre and radius $\overline{OA}$, draw arc of one complete revolution. [it is shown in the fig (i)]

Step 3. This is the required circle of radius 2 cm. [fig(ii)]

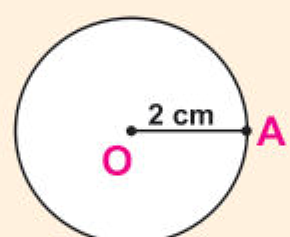


Fig (ii)

Teacher's Note Teacher may draw circle on black board and explain all these terms involving with students.